Context-Aware Compressed Sensing of Hyperspectral Image

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***Abstract*— Traditional hyperspectral imaging technique obtains numerous hyperspectral images (HSIs) with hundreds of spectral bands, leading to high cost in data acquisition, transmission, and storage. Compressed sensing (CS) theory provides a new imaging mechanism, which relies on the assumption that signals can be sparsely represented over a dictionary. By the CS imaging technique, original HSIs can be approximately reconstructed from only a few sampled measurements. In this article, a novel context-aware CS (CACS) method for HSIs is proposed by incorporating contextual prior to the dictionary learning and the sparse reconstruction. First, a patch-based online dictionary learning (ODL) algorithm is developed by introducing a joint sparse constraint. On the one hand, the online dictionary learning mechanism enables a more adaptive representation of HSIs with different scenes than using fixed-basis-based dictionaries, e.g., the discrete cosine transform (DCT) and the discrete wavelet transform dictionaries. On the other hand, the introduced joint sparse constraint promotes the learned dictionary to more sparsely and structurally represent spectral pixels. Then, with the well-learned dictionary, a weighted smoothing regularization is introduced to develop a new sparse reconstruction model. Considering the high spectral–spatial similarity of pixels in a neighborhood, the new sparse reconstruction model will encourage a locally smoothing reconstruction result. In this way, the spectral–spatial structures of the HSI can be well preserved, while possible artifacts can be effectively reduced. Experimental results demonstrate the superiority of the proposed method over some state-of-the-art hyperspectral compressive imaging methods.**

***Index Terms*— Compressed sensing (CS), contextual prior, hyperspectral image (HSI), online dictionary learning (ODL), sparse reconstruction.**

I. Introduction

HYPERSPECTRAL imaging is a crucial and powerful tool to quantitatively analyze material substances, especially in the field of remote sensing [1]–[3]. A remote sensing hyperspectral image (HSI) is a data cube, which is generally composed of more than 100 visible, near-infrared, and shortwave infrared spectral bands [4]. Since HSIs can provide rich spectral and spatial information, they have attracted much attention in different applications, e.g., land use analysis, precision agriculture, and pollution monitoring.

However, the large volumes of hyperspectral data increase the cost of sampling, storage, and transmission, which urgently require data compression process [5]. For example, the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) sensor acquires HSI with 224 spectral channels (from 0.4 to 2.5 μm) and produces gigabytes of data per flight [6]. Traditional HSI compression methods attempt to compress hyperspectral data after obtaining the whole original data [7]. By reconstructing original images from lower dimensional measurements, the compressive imaging methods [8]–[11] sample much less data, and thus greatly alleviate the burden of data acquisition. This advantage has stimulated a considerable research interest in developing HSI compressive imaging techniques for the past few years.

Typically, the principal component analysis (PCA) [12] algorithm has been used for HSI compressive imaging. For example, the compressive-projection PCA (CPPCA) [13] method randomly projects an HSI on lower dimensional subspaces at the sensor, and then reconstructs the original HSI using the PCA reconstruction. The main principal components and their coefficients used in the PCA reconstruction are estimated by the Rayleigh–Ritz theory. Afterward, some variants [14]–[16] of the CPPCA method are developed to improve the reconstruction performance. For instance, Ly *et al*. [14] incorporate both the spectral and spatial partitions of HSIs into the CPPCA method to group correlated spectral bands for better reconstruction performance. Li *et al*. [15] propose to first classify HSI pixels into different classes, and then apply the CPPCA method to each class. In this way, individual data distribution with respect to each class can be exploited to further improve the reconstruction performance. More recently, a multi-hypothesis prediction-based CPPCA (MH-CPPCA) method [16] introduces a step of postprocessing, which employs the spatial prediction and residual reconstruction to better recover original HSIs. One limitation of these CPPCA-based methods is that they require to orthogonalize random projection matrixes, leading to a more complex sampling process. In addition, these methods rely on the Rayleigh–Ritz theory that requires well-separated eigenvalues. However, this requirement might not be satisfied in practice due to the strong correlation of spectral pixels in the HSI [17].

In addition, the spectral compressive acquisition (SpeCA) [17] and the coded aperture snapshot spectral

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imager (CASSI) [18] are another two well-known approaches for hyperspectral compressive imaging. The SpeCA method is based on the linear spectral unmixing model [19], which seriously relies on a fundamental assumption that spectral vectors live in a low-dimensional subspace. The CASSI method projects a 3-D HSI cube to a 2-D matrix with a coded aperture and then reconstructs original HSIs using some optimization methods [20]–[22]. The problem of the CASSI is that it is difficult to balance the tradeoff between the spatial and the spectral resolution, especially when the number of spectral bands is large [23]. To recover an HSI with high spectral–spatial resolution, some modified CASSI approaches [21], [22] usually fuse some correlative multi-source images in the reconstruction process.

The compressed sensing (CS) [24] is an advanced signal processing theory that can effectively reconstruct sparse or compressible signals with only a few measurements [25]. In the past few years, the CS has been successfully applied to HSIs [26]–[28] based on the assumption that spectral pixels can be sparsely represented over a well-constructed dictionary [29]–[32]. For instance, the multi-task Bayesian CS (MT-BCS) method [26] regards CS of multiple pixels as a multi-task-type process. The CS combined with the unmixing technique (named by the CSU) [27] samples lower dimensional data in the spatial domain and exploits the spectral mixing characteristics in HSIs to improve the reconstruction performance. The iterative total variation (ITV) method [28] reconstructs original HSIs from a few measurements by introducing a progressive CS reconstruction framework. In addition, Martín *et al.* [33] propose a new CS methodology for HSIs, named by hyperspectral coded aperture (HYCA). This method exploits the linear mixing model and the total variation regularization to preserve the spectral and spatial structures of the HSIs. In these CS-based methods, a good predefined dictionary is necessary to sparsify hyperspectral data. However, it is difficult to construct a predefined dictionary that can adaptively characterize different images well.

More advanced dictionary learning (DL) methods are used to improve the CS reconstruction performance, rather than using predefined dictionaries. Specifically, the DL aims at learning a data-dependent dictionary from the undersampled measurements. For example, a DL-based CS method for HSIs, named by structured sparsity-based hyperspectral blind CS (SSHBCS) [34], is proposed. Owing to the joint use of a structured sparsity prior and a structured dictionary prior in the DL model, the SSHBCS method can obtain a well-learned dictionary that sparsely represents spectral pixels. However, the SSHBCS method does not make full use of the rich spectral–spatial contextual information of the HSI, which is useful for further improving the spectral–spatial preservation capability [35].

In this article, we propose a novel context-aware CS (CACS) method for the HSIs. This method mainly contains three steps. First, the lower dimensional measurements with respect to spectral pixels are sampled by means of random measurement matrixes. Then, a patch-based online DL (ODL) algorithm is introduced to learn a dictionary from the lower dimensional measurements, by alternating the sparse decomposition and the dictionary update in each iteration. Based on the fact that neighboring pixels within a patch tend to represent the same ground materials and have high spectral–spatial correlation, thus, they can be jointly represented with a common few dictionary atoms. To exploit such spectral–spatial characteristics, a joint sparse constraint is introduced in the iterative ODL model. As a result, the dictionary is learned to promote a structural and sparser representation of spectral pixels. Finally, the widely used sparsity regularization is combined with a weighted smoothing regularization to construct a new sparse reconstruction model. Here, the introduced smoothing prior is used to drive the reconstructed neighboring pixels to be similar. Consequently, the spectral and spatial characteristics of the HSIs can be better preserved. Experimental results on real HSIs demonstrate the superiority of the proposed CACS method to some state-of-the-art hyperspectral compressive imaging methods, in terms of the average data fidelity, the spatial structure fidelity, and the spectral fidelity.

The remainder of this article is organized as follows. Section II briefly introduces related works. The details of the proposed CACS method are described in Section III. Section IV presents experimental results and discussions. Section V concludes this article and gives some possible suggestions about future works.

II. Related Works

In this section, we briefly introduce the general CS method for the HSIs. Let $\mathbf{x} \in \mathbb{R}^B$ represent a spectral pixel with B spectral values. The general CS method adopts linear measurement to sample each pixel by multiplying the pixel with a "fat" measurement matrix $\mathbf{A} \in \mathbb{R}^{q \times B} (q < B)$, as follows:

$$\mathbf{y} = \mathbf{A}\mathbf{x} \tag{1}$$

where $\mathbf{y} \in \mathbb{R}^q$ is the measurements of the CS. The sampling rate can reach to q/B. It is difficult to recover pixel $\mathbf{x}$ by means of the underdetermined system of equations in (1). In general, it is required that $\mathbf{x}$ should be sparse or can be sparsely represented over a dictionary. Considering spectral pixels are dense signals rather than sparse signals, a dictionary $\mathbf{D}$ is required to sparsify the pixel. Typically, the dictionary $\mathbf{D}$ is predefined by using some fixed orthogonal bases (e.g., discrete cosine bases [36], discrete wavelet bases [37]) or learned dictionary [38]. Then, $\mathbf{x}$ can be denoted by $\mathbf{x} = \mathbf{D}\boldsymbol{\alpha}$, where $\boldsymbol{\alpha}$ is the sparse coefficient.

To guarantee an effective reconstruction result, the product of matrixes $\mathbf{A}$ and $\mathbf{D}$, i.e., $\Psi = \mathbf{AD}$, should satisfy a restricted isometry property (RIP) condition [25], [39]. Mathematically, the RIP condition is represented as

$$(1 - \sigma_s)\|\boldsymbol{\alpha}\|_2 \leq \|\Psi\boldsymbol{\alpha}\|_2 \leq (1 + \sigma_s)\|\boldsymbol{\alpha}\|_2 \tag{2}$$

where σ_s is a restricted isometric constant and s denotes the sparsity level of $\boldsymbol{\alpha}$. With a predefined dictionary $\mathbf{D}$, a matrix $\mathbf{A}$ should be properly constructed to make Ψ satisfy the RIP condition. A commonly used way to construct $\mathbf{A}$ in practice is adopting some random matrixes, e.g., random Gaussian and Bernoulli matrixes [40].

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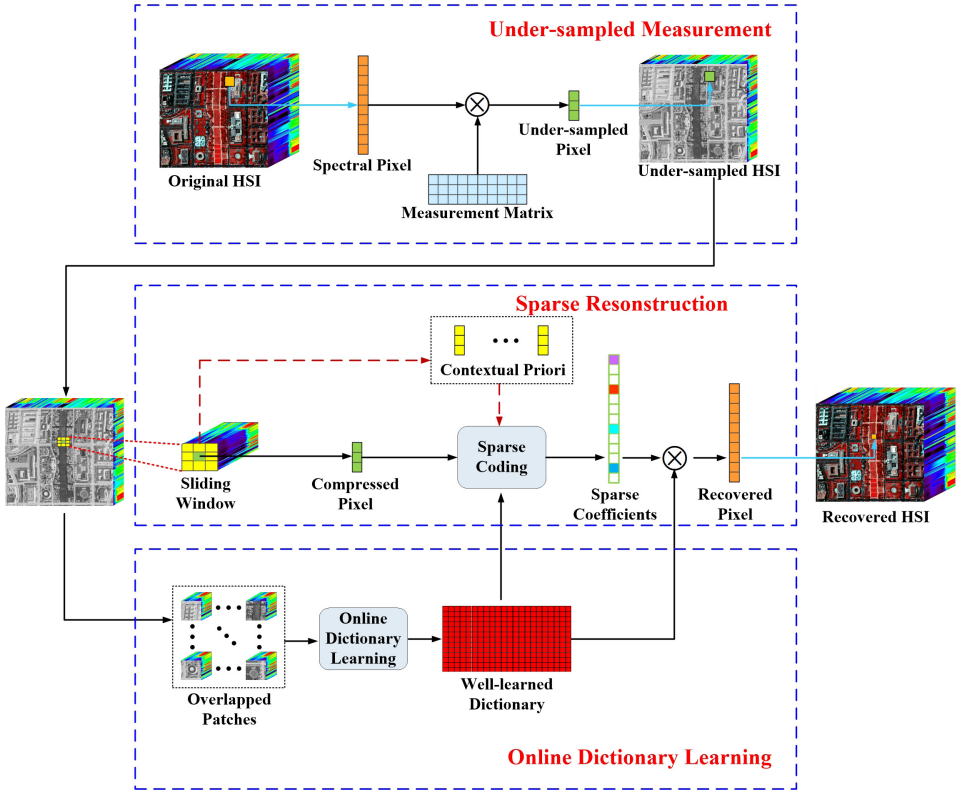


Fig. 1. Flowchart of the proposed CACS method.

With the known $\mathbf{y}$, $\mathbf{A}$, and $\mathbf{D}$, sparse coefficients $\boldsymbol{\alpha}$ can be calculated by solving the following optimization problem:

$$\min_{\boldsymbol{\alpha}} \|\boldsymbol{\alpha}\|_0 \quad \text{s.t.} \quad \|\mathbf{y} - \Psi\boldsymbol{\alpha}\|_2 < \varepsilon \tag{3}$$

where ε is a small constant. The nonconvex ℓ_0 regularization in (3) can be relaxed to a ℓ_1-type one, leading to the following transformed convex optimization problem:

$$\hat{\boldsymbol{\alpha}} = \arg\min_{\boldsymbol{\alpha}} \frac{1}{2}\|\mathbf{y} - \Psi\boldsymbol{\alpha}\|_2 + \lambda\|\boldsymbol{\alpha}\|_1. \tag{4}$$

The problem in (4) can be solved effectively with various sparse coding algorithms, e.g., the fast iterative shrinkage-thresholding algorithm [41] and the two-step iterative shrinkage/thresholding algorithms [42]. Based on the estimated sparse coefficient $\hat{\boldsymbol{\alpha}}$, the original pixel $\mathbf{x}$ can be recovered by $\hat{\mathbf{x}} = \mathbf{D}\hat{\boldsymbol{\alpha}}$.

III. Proposed Method

The flowchart of the proposed CACS method is shown in Fig. 1, which mainly contains three steps, i.e., undersampled measurement, ODL, and sparse reconstruction. For the first step, the undersampled HSI is obtained by the use of random measurement matrixes. Here, we will not detailedly describe this step, which is a commonly adopted undersampling process in the CS-based methods [27], [28], [33]. Then, the HSI is partitioned into overlapped patches in the spatial domain, which are used for ODL. Finally, sparse coefficients are calculated and employed to reconstruct the original HSI in the sparse reconstruction. Our main contributions are in Steps 2 and 3, and the related illustration and analysis will be given in detail as follows.

A. Online Dictionary Learning

Here, we would like to introduce the motivation of the proposed new ODL method first. The sparsity of signals is very important for effective CS reconstruction. However, most of the spectral values in HSIs are nonzero, which leads to dense spectral signals. Thus, a carefully designed dictionary $\mathbf{D}$ is necessary to sparsify spectral pixels. Traditional dictionaries are constructed via fixed basis [e.g., discrete cosine transform (DCT) and discrete wavelet transform (DWT)], without considering the rich spectral and spatial structures of different HSIs. In contrast, the data-driven DL methods can more adaptively characterize image details [34], [43], [44] due to the use of learning mechanism. In addition, general DL methods independently decompose each pixel into sparse coefficients without considering the contextual information. Considering that neighboring pixels have high spatial correlation and spectral similarity, they can share a similar sparsity pattern. Therefore, pixels within a local patch can be jointly represented with a few selected dictionary atoms during the DL process. In this way, spectral pixels will be represented by structural and sparser coefficients over the learned dictionary, contributing to a better reconstruction result.

Based on the above analysis, a new ODL approach is developed, which uses not only the sparsity of signals but also the implicit spectral–spatial contextual information within compressed measurements. Different from traditional DL (DL) methods [45], [46], which are operated in the original signal

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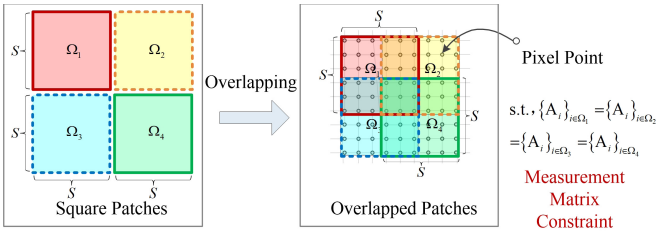


Fig. 2. Sketch map of overlapped patches used to explore contextual information of HSIs.

space, the proposed ODL aims at learning an optimized dictionary online from low-dimensional sampled data. Let $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_N] \in \mathbb{R}^{B\times N}$ denote an HSI with N spectral pixels, where B is the number of spectral bands. $\mathbf{D} = [\mathbf{d}_1, \ldots, \mathbf{d}_M] \in \mathbb{R}^{B\times M}$ represents the dictionary with M atoms. To generate a better sparse representation of measured pixels, we jointly use similar pixels to iteratively update the dictionary. As shown in Fig. 2, overlapped square regions are used to extract similar pixels instead of nonoverlapped regions to eliminate the possible artifacts. Let Ω_t represent the square region in the tth iteration. The undersampled measurements of those neighboring pixels are represented by $\{\mathbf{y}_i | \mathbf{y}_i = \mathbf{A}_i\mathbf{x}_i,\ i \in \Omega_t\}$, where $\mathbf{A}_i = \binom{\mathbf{A}_i^d}{\mathbf{A}^c}$ and $\mathbf{y}_i = \binom{\mathbf{y}_i^d}{\mathbf{y}_i^c}$. $\mathbf{A}^c$ is a random common measurement matrix for different pixels. A different measurement matrix $\mathbf{A}_i^d$ is applied to the pixel $\mathbf{x}_i$. The common measurement matrix helps to calculate sparse coefficients in each iteration. The different measurement matrixes $\{\mathbf{A}_i | i \in \Omega_t\}$ are used to avoid projecting pixels onto only one low-dimensional random subspace. Otherwise, the original signal space will be lost, which makes it difficult to learn a high-dimensional dictionary from low-dimensional data [47].

In each iteration, the ODL first sparsely decomposes input measurements while the dictionary is fixed. Then, the dictionary is updated by minimizing the error of sparse decomposition. At the beginning of the ODL, we construct an initial "undersampled dictionary" Ψ_0, by uniformly selecting some undersampled measurements from $\{\mathbf{y}_i^c | i = [1, \ldots, n]\}$. In general, the sparse decomposition model in the tth iteration can be described as

$$\arg\min_{\Phi_t} \sum_{i\in\Omega_t} \left(\frac{1}{2} \left\| \mathbf{y}_i^c - \Psi_{t-1}\boldsymbol{\alpha}_i \right\|_2^2 \right) + \psi(\Phi_t) \tag{5}$$

where $\Phi_t = \{\boldsymbol{\alpha}_i |_{i\in\Omega_t}\}$ denotes the sparse coefficient matrix and $\psi(\Phi_t)$ is a regularizer used to induce the sparsity and other prior knowledge in the model.

Based on the assumption that neighboring pixels tend to be similar, they can be approximately represented by a linear combination of a few common dictionary atoms. To use this contextual prior, the estimated coefficient matrix Φ_t is constrained to be *row-sparse*. However, it is inevitable to draw a few "outliers" due to the use of the window-based scanning mode. These "outliers" are the pixels that are very different from the majority of the rest of the pixels in a neighborhood. If these "outliers" are imposed to share the same sparsity pattern, the acquisition of latent sparse subspace will be degraded. With this consideration, we hope to find the outliers and exclude them when updating the dictionary.

Algorithm 1 AM Method to Solve (7)

Input: set $k = 0$, $t_0 = 1$; λ_1, λ_2, $\eta > 0$; $\mathbf{P}^0$, $\mathbf{Q}^0$;
set $\mathbf{R}^1 = \mathbf{P}^0$, $\mathbf{Z}^1 = \mathbf{Q}^0$;
Output: solution of $\mathbf{P}$ and $\mathbf{Q}$

1: **repeat**
2: $\mathbf{P}^{k+1} \leftarrow \arg\min_{\mathbf{P}} \frac{1}{2}\|\mathbf{P} - (\mathbf{R}^k - \frac{1}{\eta}\nabla_{\mathbf{R}}(\mathbf{R}^k, \mathbf{Z}^k))\|_F^2 + \frac{\lambda_1}{\eta}\|\mathbf{P}\|_{1,2}$
3: $\mathbf{Q}^{k+1} \leftarrow \arg\min_{\mathbf{Q}} \frac{1}{2}\|\mathbf{Q} - (\mathbf{Z}^k - \frac{1}{\eta}\nabla_{\mathbf{Z}}(\mathbf{R}^k, \mathbf{Z}^k))\|_F^2 + \frac{\lambda_2}{\eta}\|\mathbf{Q}^{\mathbf{T}}\|_{1,2}$
4: $t_{k+1} \leftarrow (1 + \sqrt{t_k^2 + 1})/2$
5: $\mathbf{R}^{k+1} \leftarrow \mathbf{P}^k + \frac{(t_{k+1}-1)}{t_k}(\mathbf{P}^k - \mathbf{P}^{k-1})$
6: $\mathbf{Z}^{k+1} \leftarrow \mathbf{Q}^k + \frac{(t_{k+1}-1)}{t_k}(\mathbf{Q}^k - \mathbf{Q}^{k-1})$
7: $k = k + 1$
8: **until** stopping criterion is satisfied

To this end, we design a mixed norm for the regularizer $\psi(\Phi_t)$ as $\psi(\Phi_t) = \lambda_1\|\mathbf{P}_t\|_{1,2} + \lambda_2\|\mathbf{Q}_t^T\|_{1,2}$, where $\|\cdot\|_{1,2}$ norm is used to induce *row-sparse* coefficients, $\mathbf{P}_t$ is *row-sparse*, and $\mathbf{Q}_t$ is *column-sparse* whose nonzero columns are corresponding to outliers. In this case, we integrate the utilization of contextual prior and outlier detection into a unified model. As a consequence, (5) becomes

$$\arg\min_{\mathbf{P}_t,\mathbf{Q}_t} \frac{1}{2}\left\|\mathbf{Y}_{\Omega_t}^c - \Psi_{t-1}(\mathbf{P}_t + \mathbf{Q}_t)\right\|_F^2 + \lambda_1\|\mathbf{P}_t\|_{1,2} + \lambda_2\left\|\mathbf{Q}_t^T\right\|_{1,2}. \tag{6}$$

To solve (6), we propose to employ the alternating minimization (AM) method (i.e., alternately optimizing each variable and fixing other variables) with an introduced proximal operator (the first-order Taylor expansion). At the kth iteration, the solution to (6) is generated by solving a surrogate function, as shown below [subscripts shown in (6) are ignored to simplify the description]

$$\begin{aligned}(\mathbf{P}^k, \mathbf{Q}^k) = \arg\min_{\mathbf{P},\mathbf{Q}} & l(\mathbf{R}, \mathbf{Z}) + \left\langle \frac{\partial l(\mathbf{R}, \mathbf{Z})}{\partial \mathbf{R}}, \mathbf{P} - \mathbf{R} \right\rangle \\ & + \frac{\eta}{2}\|\mathbf{P} - \mathbf{R}\|_F^2 + \left\langle \frac{\partial l(\mathbf{R}, \mathbf{Z})}{\partial \mathbf{Z}}, \mathbf{Q} - \mathbf{Z} \right\rangle \\ & + \frac{\eta}{2}\|\mathbf{Q} - \mathbf{Z}\|_F^2 + \psi(\Phi)\end{aligned} \tag{7}$$

where $l(\mathbf{R}, \mathbf{Z}) = (1/2)\|\mathbf{Y} - \Psi(\mathbf{R}_t + \mathbf{Q}_t)\|_F^2$, and the first five terms are the Taylor expansion of $l(\mathbf{P}, \mathbf{Q})$ at $(\mathbf{R}^k, \mathbf{Z}^k)$. Then, different variables are alternately updated until convergence. The detailed description of the AM method to solve (7) is demonstrated in Algorithm 1. Note that $\nabla_{\mathbf{R}}(\mathbf{R}^k, \mathbf{Z}^k)$ and $\nabla_{\mathbf{Z}}(\mathbf{R}^k, \mathbf{Z}^k)$ are the partial derivatives of $l(\mathbf{R}, \mathbf{Z})$. The optimization problems of Steps 2 and 3 in Algorithm 1 can be solved by iterative shrinkage-thresholding algorithms [41], [42].

After sparse coefficients are obtained, outliers are detected and removed from dictionary update, while the remaining k_t samples in the tth iteration. Let $\mathbf{y}_{i,t}$ represent the compressed measurement with respect to the ith pixel drawn in the tth iteration. $\mathbf{A}_{i,t}$ and $\boldsymbol{\alpha}_{i,t}$ are the corresponding measurement matrix

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Algorithm 2 BCD Method to Solve (9)

Input: $\mathbf{Y}_\tau$, $\mathbf{K}_\tau$; $\hat{\mathbf{D}}_{t-1}$; $U = \sum_{\tau=1}^{t} \mathbf{K}_\tau^T \mathbf{K}_\tau$; $V = \sum_{\tau=1}^{t} \mathbf{K}_\tau^T vec(\mathbf{Y}_\tau)$;
Output: dictionary $\hat{\mathbf{D}}_t \in \mathbb{R}^{B\times M}$

Set $\mathbf{D} = \hat{\mathbf{D}}_{t-1}$;
repeat
  for $m = 1$ to M **do**
    Update the mth atom $\mathbf{d}_m$ by optimizing:

$$\arg\min_{\mathbf{d}_m} \frac{1}{t}\sum_{\tau=1}^{t}\frac{1}{2}\|vec(\mathbf{Y}_\tau) - \mathbf{K}_{\tau,-m}vec(\mathbf{D})_{-m} - \mathbf{K}_{\tau,m}\mathbf{d}_m\|_2^2$$

    Using atoms in last iteration as the warm start:

$$\mathbf{d}_m \leftarrow \mathbf{d}_m + \left(\sum_{\tau=1}^{t}\mathbf{K}_{\tau,m}^T\mathbf{K}_{\tau,m}\right)^{-1} \cdot \left(\sum_{\tau=1}^{t}\mathbf{K}_{\tau,m}^T vec(\mathbf{Y}_\tau) - \sum_{\tau=1}^{t}\mathbf{K}_{\tau,m}^T\mathbf{K}_\tau\mathbf{D}\right)$$

    where $\sum_{\tau=1}^{t}\mathbf{K}_{\tau,m}^T\mathbf{K}_{\tau,m}$, $\sum_{\tau=1}^{t}\mathbf{K}_{\tau,m}^T vec(\mathbf{Y}_\tau)$ are submatrixes of U and V, respectively.
    Normalization: $\mathbf{d}_m \leftarrow \frac{\mathbf{d}_m}{\max(1,\ \|\mathbf{d}_m\|_2)}$
  end for
until convergence
Return $\hat{\mathbf{D}}_t = \mathbf{D}$

and sparse coefficients. With the consideration of learned information in previous iterations, the dictionary update model in the tth iteration, which fixes sparse coefficients and optimizes $\mathbf{D}$, can be represented as

$$\arg\min_{\mathbf{D}} \frac{1}{t}\sum_{\tau=1}^{t}\sum_{i=1}^{k_\tau}\frac{1}{2}\|\mathbf{y}_{i,\tau} - \mathbf{A}_{i,\tau}\mathbf{D}\boldsymbol{\alpha}_{i,\tau}\|_2^2. \tag{8}$$

Equation (8) can be rewritten in the form of *Kronecker* operator [48] as

$$\hat{\mathbf{D}}_t = \arg\min_{\mathbf{D}} \frac{1}{t}\sum_{\tau=1}^{t}\frac{1}{2}\|\mathrm{vec}(\mathbf{Y}_\tau) - \mathbf{K}_\tau\mathrm{vec}(\mathbf{D})\|_2^2 \tag{9}$$

where $\mathbf{K}_\tau = [(\boldsymbol{\alpha}_{1,\tau}\otimes\mathbf{A}_{1,\tau})^T, \ldots, (\boldsymbol{\alpha}_{k_\tau,\tau}\otimes\mathbf{A}_{k_\tau,\tau}^T)]^T$, $\mathbf{Y}_\tau = [\mathbf{y}_{1,\tau}, \ldots, \mathbf{y}_{k_\tau,\tau}]$, vec$(\cdot)$ is the vectorization operator, and $\otimes$ means the *Kronecker* product. Since the coefficients of most atoms are zero in each iteration, only a few atoms are selected to be updated. Therefore, a modified block-coordinate descent (BCD) algorithm is developed to solve (9), i.e., alternately updating each atom while keeping other atoms fixed. This algorithm is summarized in Algorithm 2, where $\mathbf{K}_{\tau,-m}$ and $\mathbf{K}_{\tau,m}$ are the submatrixes after removing/extracting the columns corresponding to $\mathbf{d}_m$ from $\mathbf{K}_\tau$, respectively. Before the next iteration, we update the "compressed dictionary" using $\hat{\mathbf{D}}_t$, i.e., $\Psi_t = \mathbf{A}^c\hat{\mathbf{D}}_t$, and then repeat the above operations.

B. Sparse Reconstruction

In the sparse reconstruction, sparse coefficients of each spectral pixel are first estimated by using the sparse coding process. The reconstruction of original pixels is then achieved by multiplying the learned dictionary by sparse coefficients. In (4), general sparse coding regards each spectral pixel as an independent signal, while the spatial correlation of the pixels is ignored. Not only the sparsity but also the contextual characteristics of the HSIs should be collaboratively used to recover the HSIs. In this way, some badly reconstructed pixels obtained by using only the sparsity of the pixels may be corrected with some contextual prior. To this end, a weighted smoothing regularization, which tries to push reconstructed neighboring pixels to be similar, is introduced in the sparse coding.

After the well-learned dictionary $\hat{\mathbf{D}}$ is obtained, the sparse coefficient of each pixel can be calculated. Instead of using only the sparsity (i.e., ℓ_1 constraint for $\boldsymbol{\alpha}$) of spectral pixels in (4), the contextual correlation of pixels is also considered. An additional constraint, which minimizes the Euclidean distances between the recovered pixel and its neighboring pixels, is applied to the general ℓ_1 model. Then, the proposed model can be represented by

$$\min_{\boldsymbol{\alpha}_1,\ldots\boldsymbol{\alpha}_n}\sum_{i=1}^{n}\left(\|\mathbf{y}_i - \mathbf{A}_i\hat{\mathbf{D}}\boldsymbol{\alpha}_i\|_2^2 + \gamma\|\boldsymbol{\alpha}_i\|_1\right) + \sum_{i\neq j\in[1,\ldots,n]}\beta\left(W_{i,j}\|\hat{\mathbf{D}}\boldsymbol{\alpha}_i - \hat{\mathbf{D}}\boldsymbol{\alpha}_j\|_2^2\right) \tag{10}$$

where $W_{i,j}$ is the weight, and $\boldsymbol{\alpha}_i$ and $\boldsymbol{\alpha}_j$ are the sparse coefficients of $\mathbf{y}_i$ and $\mathbf{y}_j$, respectively. Let (r_i, c_i) denote the spatial coordinates of the pixel with respect to $\boldsymbol{\alpha}_i$. Here, the weight is calculated according to the spatial distance $S_{i,j} = (r_i - r_j)^2 + (c_i - c_j)^2$ of two pixels with a Gaussian function

$$W_{i,j} = \begin{cases} a\cdot e^{-S_{i,j}/2\sigma^2}, & (S_{i,j}\leq 2) \\ 0, & (S_{i,j} > 2) \end{cases} \tag{11}$$

where $\sigma = 0.8$ and a is chosen to satisfy $\sum_j W_{i,j} = 1, \forall i \in [1,\ldots,n]$. To solve the optimization problem in (11), we iteratively optimize each $\boldsymbol{\alpha}_i$ until convergence. That is, all the remaining sparse coefficients $\boldsymbol{\alpha}_j (i \neq j)$ are fixed, when $\boldsymbol{\alpha}_i$ is optimally updated. We can rewrite (10) with respect to $\boldsymbol{\alpha}_i$ as follows:

$$\min_{\boldsymbol{\alpha}_i} J(\boldsymbol{\alpha}_i) + \gamma\|\boldsymbol{\alpha}_i\|_1 \tag{12}$$

where

$$J(\boldsymbol{\alpha}_i) = \|\mathbf{y}_i - \mathbf{A}_i\hat{\mathbf{D}}\boldsymbol{\alpha}_i\|_2^2 + \beta\sum_{j=1}^{n}W_{i,j}\|\hat{\mathbf{D}}\boldsymbol{\alpha}_i - \hat{\mathbf{D}}\boldsymbol{\alpha}_j\|_2^2 \tag{13}$$

is differentiable. $\boldsymbol{\alpha}_i$ is calculated by a developed variant of the feature-sign search (FSS) algorithm [49], as summarized in Algorithm 3.

Finally, the reconstruction of the HSIs from sparse coefficients is implemented by

$$\hat{\mathbf{X}} = \hat{\mathbf{D}}\Phi \tag{14}$$

where $\Phi = \{\boldsymbol{\alpha}_1, \ldots, \boldsymbol{\alpha}_N\}$ is sparse coefficient matrix.

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Algorithm 3 FSS Method to Solve (12)

Input: dictionary $\hat{\mathbf{D}}$; measurement $\mathbf{y}_i$; matrix W; initial $\boldsymbol{\alpha}_i$; $\beta > 0$;

Output: solution of $\boldsymbol{\alpha}_i$

1: **Initialize:** $\theta = sign(\boldsymbol{\alpha}_i)$, values of θ are from $\{-1, 0, 1\}$, active set $\Lambda := \text{Find}(\boldsymbol{\alpha}_i \neq 0)$;
2: Denote: $\Gamma = \frac{\partial J(\boldsymbol{\alpha}_i)}{\partial \boldsymbol{\alpha}_i}$, $\widetilde{\Gamma} = \frac{\partial^2 J(\boldsymbol{\alpha}_i)}{\partial \boldsymbol{\alpha}_i^2}$;
3: From zero coefficients of $\boldsymbol{\alpha}_i$, select $h = \arg\min_r |\Gamma^r|$ (Γ^r denotes r^{th} entries in Γ);
4: **if** $\Gamma^h > \gamma$, **then** set $\theta_h := -1$, $\Lambda := \{h\} \cup \Lambda$;
5: **if** $\Gamma^h < -\gamma$, **then** set $\theta_h := 1$, $\Lambda := \{h\} \cup \Lambda$;
6: **Feature-sign step:**
7: Let $\hat{\mathbf{D}}_\Lambda$ and $\widetilde{\Gamma}_\Lambda$ be submatrixes of $\hat{\mathbf{D}}$ and $\widetilde{\Gamma}$ that contain only the columns corresponding to the active set Λ;
8: Let $\boldsymbol{\alpha}_\Lambda$ and θ_Λ be subvectors of $\boldsymbol{\alpha}_i$ and θ corresponding to the active set Λ;
9: Compute the analytical solution to the resulting unconstrained QP: $\min_{\boldsymbol{\alpha}_\Lambda} J(\boldsymbol{\alpha}_\Lambda) + \gamma\theta_\Lambda\boldsymbol{\alpha}_\Lambda$, by
10: $\hat{\boldsymbol{\alpha}}_\Lambda = (\widetilde{\Gamma}_\Lambda)^{-1}(2\hat{\mathbf{D}}_\Lambda^T\mathbf{A}_i^T\mathbf{y}_i + 2\beta\hat{\mathbf{D}}_\Lambda^T\sum_{j=1}^{n} W_{i,j}\hat{\mathbf{D}}\boldsymbol{\alpha}_j + \gamma\theta_\Lambda)$;
11: Perform a discrete line search on the closed line segment from $\boldsymbol{\alpha}_\Lambda$ to $\hat{\boldsymbol{\alpha}}_\Lambda$:
12: (a) Check the objective value at $\hat{\boldsymbol{\alpha}}_\Lambda$ and all points where any coefficient changes sign,
13: (b) Update $\boldsymbol{\alpha}_\Lambda$ (and the corresponding entries in $\boldsymbol{\alpha}_i$) to the point with the lowest objective value.
14: Remove zero coefficients from the active set Λ and update $\theta = sign(\boldsymbol{\alpha}_i)$;
15: **Check the optimality conditions:**
16: (a) $\Gamma^r + \gamma\, sign(\boldsymbol{\alpha}_i^r) = 0$, *s.t.*, $\forall\boldsymbol{\alpha}_i^r \neq 0$. If (a) is not satisfied, go to Step 8, else check condition (b);
17: (b) $|\Gamma^r| < \gamma$, *s.t.*, $\forall\boldsymbol{\alpha}_i^r = 0$. If (b) is not satisfied, go to Step 5, otherwise output $\boldsymbol{\alpha}_i$ as solution.

IV. Experimental Results and Discussion

In this section, three different hyperspectral data sets are used in the experiments to demonstrate the effectiveness of the proposed CACS method. The three hyperspectral data sets contain different types of scenes and are used as the original images that we are trying to reconstruct. In the experiments, each HSI is first undersampled by predefined measurement matrixes, which simulates the sampling process in the hyperspectral compressive imaging. The acquired measurements are then used for the reconstruction of the original HSI with different methods. In the experiments, the proposed method is compared with several state-of-the-art methods.

A. Data Sets

Three well-known HSIs are used in the experiments, including the Indian Pines image,[1] the Cuprite image,[1] and the Washington DC Mall image.[2]

[1] http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

[2] https://engineering.purdue.edu/~ biehl/MultiSpec/hyperspectral.html

1) Indian Pines Image: The hyperspectral data set is collected by the AVIRIS senor over the agricultural Indian Pine test site in northwestern Indiana. This data set with the spatial resolution of 20 m has 224 spectral bands across the spectral range from 0.4 to 2.5 μm. The HSI used in our experiments is a subimage of the data set and has the size of $145 \times 145 \times 200$ (24 water absorption and broken bands are removed).

2) Cuprite Image: This hyperspectral data set is a commonly used benchmark data set for the hyperspectral unmixing research and captured by the AVIRIS sensor over the Cuprite in Las Vegas, NV, USA. Owing to the sampling from the same sensor, the spectral and spatial resolution of this data set is similar to the Indian Pines data set. The HSI used in our experiment is a subimage of the data set and contains 200 spectral bands after omitting 24 water absorption and broken bands. Each band consists of 250×190 pixels.

3) Washington DC Mall Image: The hyperspectral data set is acquired by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) sensor over the Mall in Washington, DC. This data set contains 210 spectral bands across the spectral range from 0.4 to 2.5 μm. The HSI used in our experiments is a subimage of the data set and has 191 spectral bands (the 19 Bands in 1.4- and 1.9-μm region where the atmosphere is opaque are removed). Each band contains 280×307 pixels with a spatial resolution of about 2.8 m.

B. Experimental Setup

1) Methods for Comparison: The proposed method is compared with several state-of-the-art HSI compressive imaging methods, including the MH-CPPCA method [16], the ITV method [28], the HYCA method [33], and the SSHBCS method [34]. Except the MH-CPPCA method that achieves the best performance among various CPPCA variants, the other three methods are CS-based methods. The three CS-based methods are different in dictionary designing. The ITV method uses the fixed orthogonal bases to construct a dictionary. The HYCA method defines the dictionary as a collection of endmembers. The SSHBCS method learns a data-dependent dictionary from the measurements by incorporating a dictionary prior and a Laplace prior into the general DL model.

2) Parameter Setting: In the proposed CACS method, there are two important parameters for the DL, i.e., the patch size $s \times s$ and the dictionary size M (the number of atoms). In this article, the patch size is empirically set to 40×40 for all the three data sets, i.e., $s = 40$. The dictionary size M is set to two times of the spectral band number B for each hyperspectral data set. The discussion about these parameters will be demonstrated in Section IV-C. In addition, the parameters of the sparse coding model, including the γ and β, are empirically chosen. Specifically, $\gamma = 50$ is applied to the three hyperspectral data sets. Owing to different dynamic ranges of the three hyperspectral data sets, β for the Indian Pines image, the Cuprite image, and the Washington DC Mall image is set to 0.1, 0.01, and 0.005, respectively. Note that the elements of the measurement matrix $\mathbf{A}_i$ is drawn as random ± 1 variables with an i.i.d uniform Bernoulli distribution. This type of measurement matrix can be easily implemented using a digital micromirror device (DMD) [40] and is applied to all the

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TABLE I

PSNR, SSIM, AND SAD VALUES OBTAINED BY THE MH-CPPCA [16], ITV [28], HYCA [33], SSHBCS [34], AND CACS METHODS OVER THE INDIAN PINES IMAGE (THE BEST OBJECTIVE METRIC VALUES FOR EACH CASE ARE GIVEN IN BOLD)

	PSNR(db)							
Methods	$q=6$	$q=8$	$q=10$	$q=12$	$q=14$	$q=16$	$q=18$	$q=20$
MH-CPPCA	15.625	18.632	28.651	32.362	34.023	34.865	35.505	36.093
ITV	26.619	27.669	28.527	29.261	29.870	30.412	30.887	31.353
HYCA	32.610	33.595	34.163	35.372	36.141	35.838	36.021	37.663
SSHBCS	20.791	24.263	24.824	24.911	24.936	25.048	25.433	27.632
CACS	**35.923**	**36.666**	**37.332**	**37.895**	**38.450**	**38.727**	**38.924**	**39.308**
	SSIM							
Methods	$q=6$	$q=8$	$q=10$	$q=12$	$q=14$	$q=16$	$q=18$	$q=20$
MH-CPPCA	0.051	0.106	0.480	0.673	0.751	0.789	0.815	0.839
ITV	0.562	0.608	0.642	0.671	0.693	0.711	0.727	0.742
HYCA	0.862	0.869	0.888	0.897	**0.917**	0.895	0.899	0.918
SSHBC-S	0.758	0.766	0.813	0.819	0.829	0.821	0.869	0.873
CACS	**0.871**	**0.884**	**0.901**	**0.912**	0.915	**0.920**	**0.923**	**0.932**
	SAD(rad)							
Methods	$q=6$	$q=8$	$q=10$	$q=12$	$q=14$	$q=16$	$q=18$	$q=20$
MH-CPPCA	0.461	0.299	0.101	0.068	0.056	0.052	0.049	0.046
ITV	0.138	0.122	0.111	0.101	0.095	0.089	0.084	0.080
HYCA	**0.044**	0.043	0.040	0.038	0.038	0.046	0.044	0.038
SSHBCS	0.174	0.159	0.160	0.156	0.157	0.159	0.137	0.112
CACS	0.045	**0.041**	**0.039**	**0.037**	**0.035**	**0.033**	**0.032**	**0.031**

other methods for a reasonable comparison. The parameters of the compared methods, including the MH-CPPCA method, the ITV method, the HYCA method, and the SSHBCS method, are set as reported in [16], [28], [33], and [34], respectively. The sampling rate used here is represented by the number of sampled spectral bands. That is, the sampling rate varies in a range of q/B, $(q = 6, 8, 10, 12, 14, 16, 18, 20)$, to more completely evaluate different compressive imaging methods.

3) Evaluation Metrics: Three different metrics are used in the experiments to evaluate the results of different methods, i.e., the peak signal-to-noise ratio (PSNR), the structure similarity (SSIM), and the spectral angle distance (SAD) [34]. Specifically, the PSNR measures the average numerical similarity between the reconstructed HSI and the original one. The SSIM is a mean value obtained by averaging the SSIM values of all bands and is used for the evaluation of spatial structure preservation. The SAD is a mean value obtained by averaging the SAD values of all pixels and is used to measure the average spectral fidelity. Note that larger PSNR, SSIM, and smaller SAD values mean better reconstruction performance.

C. Experiments With Indian Pines Image

In Table I, the results of different HSI imaging methods on the Indian Pines data set are objectively evaluated. It can be observed that the PSNR of the proposed CACS method steadily rises as the sampled band number q increases. When compared with other methods, the proposed CACS method consistently obtains the highest PSNR with different sampling rates. The average gain of the PSNR by the proposed CACS method is about 2 dB, which demonstrates the lower distortion rate of the CACS method for image reconstruction. In terms of the SSIM and SAD metrics, the CACS method has the best values in most cases, indicating a comparable performance with respect to the spectral shape and spatial structure fidelity for the Indian Pines image. It should be noted that the HYCA method uses a good predefined dictionary (i.e., endmember matrix), and thus shows competitive performance in SAD metric.

Fig. 3 shows one of the visual results (the tenth band) with a low sampling rate (i.e., $q = 6$). Here, the corresponding residual images between the reconstructed bands and the original bands are also given for a visual comparison. It can be observed that the reconstructed band of the MH-CPPCA method is much noisy, which is in accordance with the lowest SSIM value in Table I. The reconstructed result by the ITV method is oversmoothing, which loses a lot of structured details, as shown in the corresponding residual image. For the HYCA method, some artifacts are introduced in the case of a low sampling rate, due to the use of 2×2 scanning window in the reconstruction algorithm. As can be seen in Fig. 3(e), it seems that the reconstructed band by the SSHBCS method has good edge and texture preservation result. However, according to the residual images obtained by the difference between this reconstructed band and the original band, a lot of points in the residual image have very large values, which means that the reconstructed band is largely different from the original band. Moreover, a different visual impression is presented by the SSHBCS method, since the spectra of HSI pixels are not well reconstructed (reflected by the lower SAD). The residual image in Fig. 3(f) shows that our CACS method acquires closer band values to the original data set, demonstrating a superior reconstruction performance.

To illustrate the effectiveness of the proposed DL algorithm in improving the reconstruction result, we conduct experiments by replacing the learned dictionary with a fixed basis for the CACS method and keeping other configurations unchanged in the meanwhile. In this experiment, the classical DCT and DWT dictionaries are used by the CACS, which are named DCT-CACS and DWT-CACS, respectively. Correspondingly,

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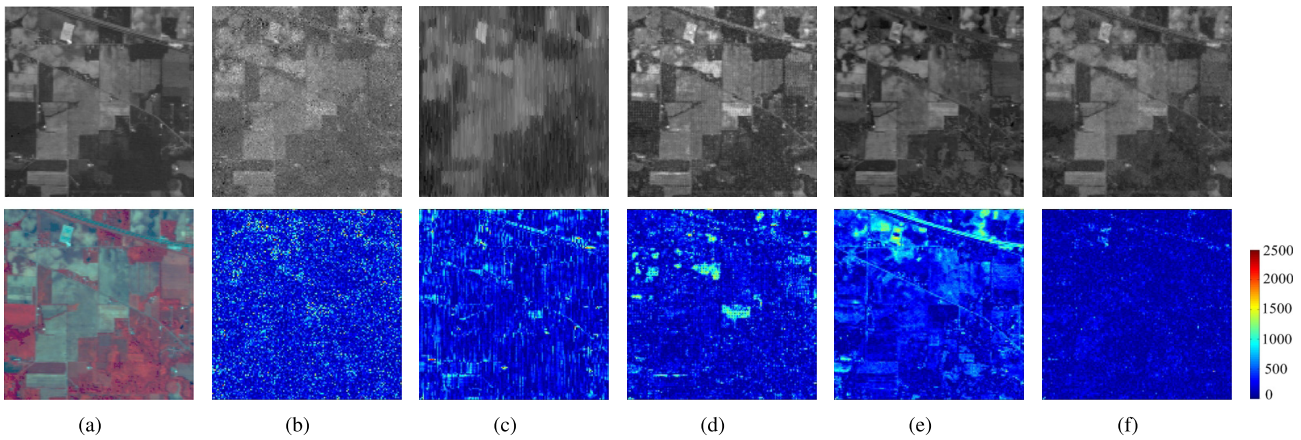


Fig. 3. Reconstructed results of the tenth band and the corresponding residual images by different methods. (a) Original tenth band and false color composite. (b)–(f) Reconstructed results by different methods, including MH-CPPCA [16], ITV [28], HYCA [33], SSHBCS [34], and CACS methods, respectively.

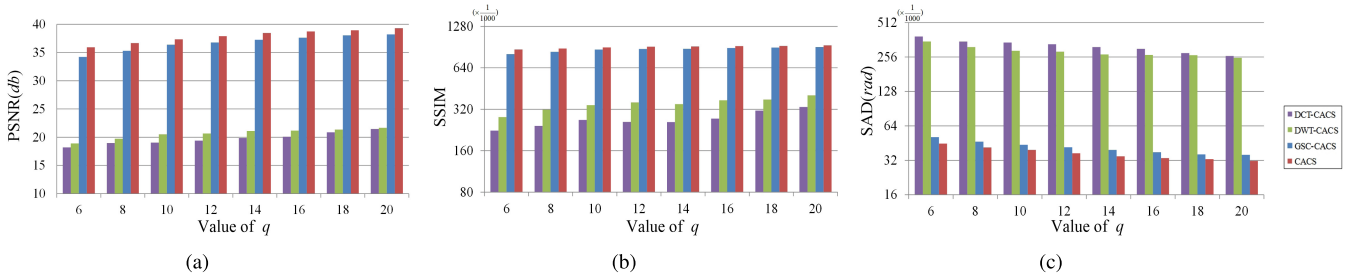


Fig. 4. Reconstruction results of the DCT-CACS, DWT-CACS, GSC-CACS, and CACS methods of the Indian Pines image with respect to different values of q, in terms of (a) PSNR, (b) SSIM, and (c) SAD.

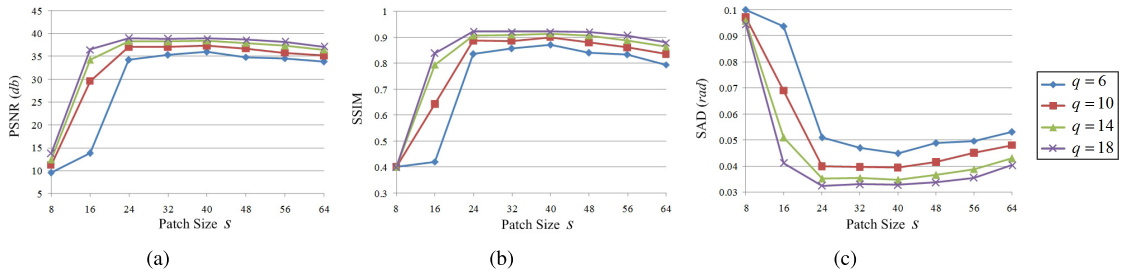


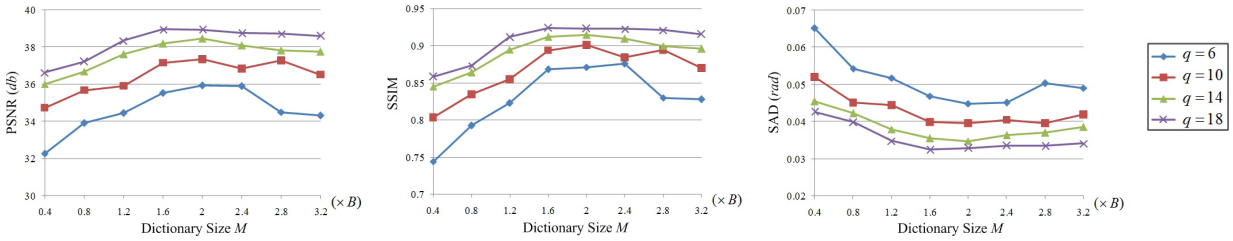
Fig. 5. Reconstruction results of the CACS method with respect to different values of patch size s, in terms of (a) PSNR, (b) SSIM, and (c) SAD.

the PSNR, SSIM, and SAD with respect to different sampling rates (controlled by q) are used to evaluate the reconstruction performance. As shown in Fig. 4, the results in terms of three objective metrics by the CACS method are much better than those of the DCT-CACS and DWT-CACS methods. It demonstrates the superiority of the CACS method in different aspects, i.e., overall data reconstruction rate, and spatial and spectral structure fidelity. This should owe to the following two reasons. One is that the online dictionary can more adaptively characterize the images of different scenes than the fixed basis. Another one is that the introduced joint sparse constraint in the DL helps for a structural and sparser representation of signals, further contributing to a better reconstruction performance.

As shown in Fig. 4, we also conduct comparison experiments to demonstrate the superiority of the proposed new sparse coding algorithm in (12) over the general sparse coding in (4). Here, the version of the CACS method by replacing (12) with (4) is named by general-spare-coding-CACS (GSC-CACS). It can be observed from Fig. 4 that, with varying values of q, the CACS method consistently outperforms the GSC-CACS method in terms of all the considered objective metrics. This demonstrates that the developed new sparse coding algorithm is meaningful for a better HSI CS reconstruction result. The main reason is that the introduction of weighted smoothing regularization in the sparse reconstruction model contributes to improve reconstruction performance in terms of spectral-spatial structures.

Moreover, we analyze the effect of important parameters on the CACS method, including the patch size s and the dictionary size M, as shown in Figs. 5 and 6, respectively. First, the effect of patch size s is discussed by varying the value of s from 12 to 70. Meanwhile, we keep other parameter

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(a) (b) (c)

Fig. 6. Reconstruction results of the CACS method with respect to different dictionary sizes M, in terms of (a) PSNR, (b) SSIM, and (c) SAD.

TABLE II

PSNR, SSIM, AND SAD VALUES OBTAINED BY THE MH-CPPCA [16], ITV [28], HYCA [33], SSHBCS [34], AND CACS METHODS OVER THE CUPRITE AND WASHINGTON DC MALL IMAGES (THE OPTIMAL VALUES FOR EACH CASE ARE GIVEN IN BOLD)

Data Set	PSNR(db)								
	Methods	$q=6$	$q=8$	$q=10$	$q=12$	$q=14$	$q=16$	$q=18$	$q=20$
Cuprite	MH-CPPCA	13.085	21.527	28.47	31.794	35.650	41.695	42.316	42.942
	ITV	28.797	29.851	30.757	31.514	32.194	32.773	33.327	33.804
	HYCA	33.360	37.213	38.104	40.930	38.505	39.405	43.565	44.880
	SSHBCS	20.386	21.316	22.879	24.026	24.240	24.846	26.562	29.975
	CACS	**37.925**	**41.846**	**42.427**	**43.206**	**43.794**	**44.747**	**45.258**	**46.075**
Washington DC Mall	MH-CPPCA	26.062	32.022	34.767	40.325	45.187	47.103	48.582	49.645
	ITV	28.848	29.723	30.470	31.131	31.735	32.287	32.806	33.304
	HYCA	36.088	39.361	42.585	45.013	46.460	45.872	48.667	50.238
	SSHBCS	23.208	24.456	25.465	25.806	26.019	26.483	27.222	29.340
	CACS	**42.166**	**44.846**	**47.221**	**48.272**	**50.011**	**50.894**	**51.788**	**52.402**
Data Set	SSIM								
	Methods	$q=6$	$q=8$	$q=10$	$q=12$	$q=14$	$q=16$	$q=18$	$q=20$
Cuprite	MH-CPPCA	0.038	0.200	0.536	0.705	0.849	0.946	0.953	0.958
	ITV	0.470	0.520	0.559	0.595	0.626	0.652	0.677	0.698
	HYCA	0.812	0.871	0.892	0.925	0.886	0.902	0.956	0.964
	SSHBCS	0.844	0.844	0.864	0.869	0.870	0.883	0.903	0.936
	CACS	**0.895**	**0.947**	**0.955**	**0.958**	**0.962**	**0.969**	**0.972**	**0.976**
Washington DC Mall	MH-CPPCA	0.295	0.509	0.624	0.811	0.908	0.942	0.958	0.968
	ITV	0.425	0.425	0.464	0.498	0.528	0.555	0.580	0.602
	HYCA	0.882	0.933	**0.956**	**0.963**	**0.975**	0.969	**0.982**	0.982
	SSHBCS	0.678	0.710	0.748	0.753	0.742	0.778	0.807	0.837
	CACS	**0.895**	**0.943**	0.951	0.962	**0.975**	**0.978**	**0.982**	**0.983**
Data Set	SAD(rad)								
	Methods	$q=6$	$q=8$	$q=10$	$q=12$	$q=14$	$q=16$	$q=18$	$q=20$
Cuprite	MH-CPPCA	0.504	0.181	0.079	0.056	0.035	0.018	0.017	0.016
	ITV	0.067	0.060	0.054	0.050	0.046	0.043	0.040	0.038
	HYCA	0.038	0.032	0.028	0.020	0.026	0.022	0.016	0.013
	SSHBCS	0.166	0.151	0.148	0.146	0.146	0.142	0.102	0.073
	CACS	**0.023**	**0.018**	**0.016**	**0.016**	**0.015**	**0.013**	**0.013**	**0.011**
Washington DC Mall	MH-CPPCA	0.367	0.172	0.122	0.063	0.038	0.029	0.025	0.022
	ITV	0.224	0.204	0.188	0.175	0.164	0.155	0.147	0.140
	HYCA	0.072	0.051	0.037	0.039	0.034	0.037	0.025	0.021
	SSHBCS	0.316	0.405	0.326	0.310	0.300	0.297	0.295	0.238
	CACS	**0.045**	**0.034**	**0.029**	**0.024**	**0.021**	**0.019**	**0.018**	**0.017**

configurations unchanged. It can be observed from Fig. 5 that the curves related to PSNR and SSIM metrics present a rise trend first along with the increase of s and then keep a relatively stable trend before a gradual drop. The SAD curves of the CACS method show an opposite trend. This is reasonable since lower SAD indicates better performance of spectral fidelity. In addition, another observation is that a range of s from 24 to 48 can lead to the approximately best performance of the CACS method. Much smaller patch size s may contain insufficient spectral–spatial information to effectively learn a good dictionary. On the contrary, too large patch size s will introduce more outliers in the DL model, which breaks the assumption that the input neighboring pixels are highly similar. As a result, the performance of the CACS method gradually drops when the patch size is set to be too small or too large.

For the dictionary size M, the value is ranged from $0.4 \times B$ to $3.2 \times B$ in the experiment where other configurations are fixed. The obtained values of the three objective metrics with respect to different values of M are shown in Fig. 6. It can be observed that a small value of M ($M \leq 1.6 \times B$) will lead to degraded performance of the CACS method. This is

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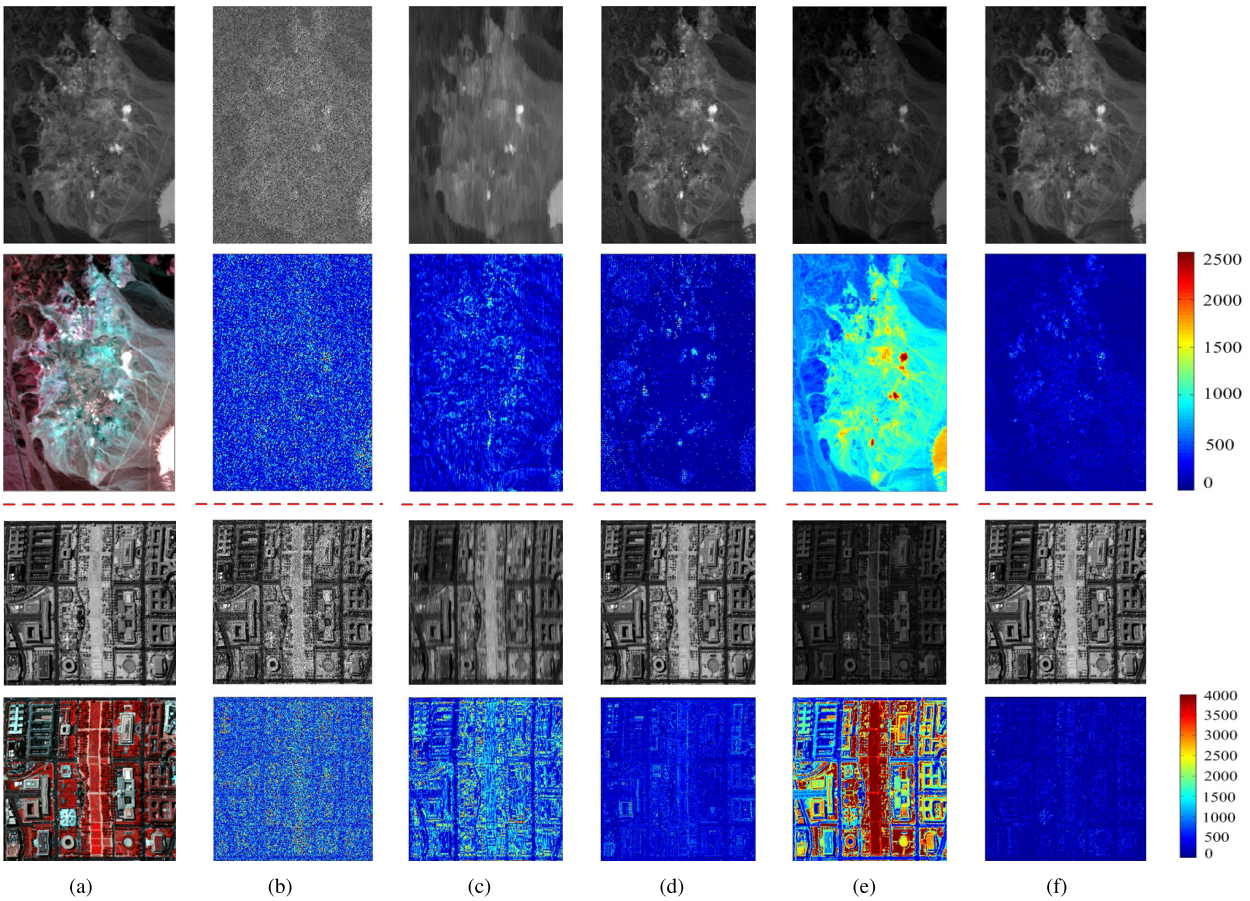


Fig. 7. Reconstructed 60th bands by different methods and corresponding residual images for the Cuprite image. Reconstructed 80th bands and residual images for the Washington DC Mall image. (a) Original 60th/80th bands and false-color composites of the Cuprite/Washington DC Mall image. (b) MH-CPPCA method [16]. (c) ITV method [28]. (d) HYCA method [33]. (e) SSHBCS method [34]. (f) CACS method.

because a small dictionary may not contain sufficient atoms for accurate and sparse representation. In addition, as the value of M increases, the performance of the CACS method remains relatively stable, especially in the cases of high sampling rate (i.e., large value of q). Therefore, an acceptable range of M is from $1.6 \times B$ to $2.4 \times B$, to make the CACS method work better with a compact dictionary.

D. Experiments With Cuprite and Washington DC Mall Images

In addition to the Indian Pines image, we also use another two widely used HSIs for experiments, i.e., the Cuprite and Washington DC Mall images. Different from the Indian Pines image, which mainly covers the agricultural areas and contains regions with regular shapes, the Cuprite and Washington DC Mall images cover the complex mine areas and urban areas, respectively. The reconstruction performance by different methods in terms of both the objective metrics and the visual impression is shown and analyzed as follows.

The objective metrics of these two data sets are shown in Table II. Among all the compared methods, it should be noted that the proposed CACS method constantly ranks the first in terms of PSNR and SAD metrics and the results of SSIM metrics are very competitive. When q is very small, the objective results for both images are not very good. The reason is that low-dimensional data may not provide enough information for accurately pursing sparse coefficients. Then, when as q increases (before $q = 14$), the results increase fast. It means that more information is sampled and the sparse coefficients can be better calculated. Finally, when the value of q exceeds 14, the PSNR slowly increases. This is because enough information is sampled for pursing sparse coefficients when $q = 14$. Moreover, according to the comparison of results in Tables I and II, it can be observed that there is a more obvious superiority in reconstructing these two data sets, especially in the cases of very low sampling rate. Taking the data set of Washington DC Mall with $q = 6$ for example, the PSNR value obtained by the CACS method is PSNR $=$ 42.166 dB, while the PSNR by the HYCA method (ranks the second) is about 6 dB lower, i.e., PSNR $=$ 36.088 dB. This is because these two images have much richer structured details than the Indian Pines data set, which is more difficult to be well reconstructed. Without taking full advantage of the spectral–spatial structure prior in both the processes of dictionary construction and sparse coding, it is difficult for the compared methods to generate good reconstruction performance. In contrast, a spectral–spatial joint sparse constraint is introduced for learning a compact dictionary to better sparsify dense spectral signals. Moreover, a weighted smoothing

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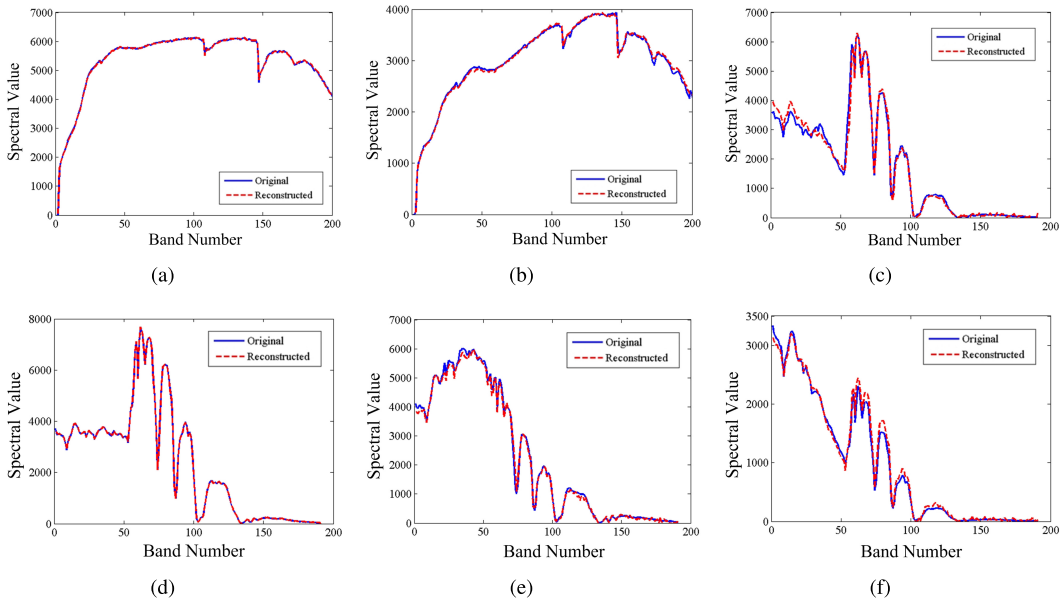


Fig. 8. Comparison results of reconstructed pixels by the CACS method and the corresponding original pixels. (a)–(c) Good case, average case, and bad case of the Cuprite image, respectively. (d)–(f) Good case, average case, and bad case of the Washington DC Mall image, respectively.

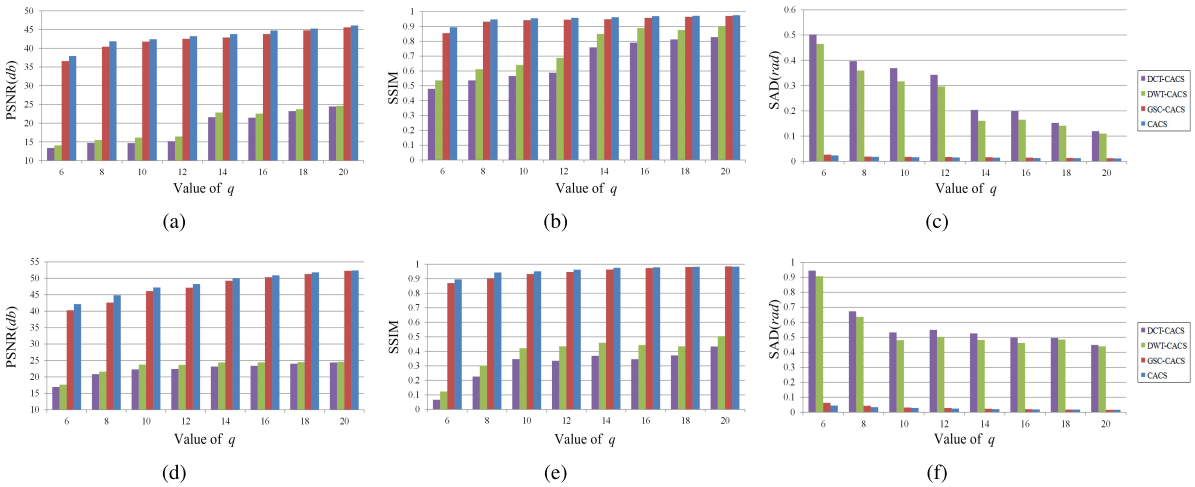


Fig. 9. Reconstruction results of the DCT-CACS, DWT-CACS, GSC-CACS, and CACS method over the Cuprite and Washington DC Mall images with respect to different values of q. (a)–(c) Reconstruction results of the Cuprite image in terms of PSNR, SSIM, and SAD, respectively. (d)–(f) Reconstruction results of Washington DC Mall image in terms of PSNR, SSIM, and SAD, respectively.

regularization is added to modify the sparse coding model, leading to a better spectral structure preservation capability. The better reconstruction performance in terms of spatial information preservation by the proposed CACS method can also be demonstrated from the visual impression in Fig. 7, where less reconstruction error (i.e., smaller values in the residual image) is obtained.

To better confirm the effectiveness of the proposed CACS method with respect to spectral fidelity, we show some visual examples of reconstructed spectral curves picked up from the Cuprite image and the Washington DC Mall image, as shown in Fig. 8. To save space, only six reconstructed spectral curves obtained in the case of $q = 12$ (i.e., relatively low sampling rate) are shown here. Specifically, we select three pixels per image, which are the representatives and correspond to the low, mean, and high SADs, respectively. Within this figure, the reconstructed pixel, which is randomly picked up from top 5% pixels with the lowest SAD, is called the "good case" here, and vice versa. The reconstructed pixel whose SAD is very close to the mean SAD is called the "average case." The blue line in each subfigure is the original spectral curve, and the red dotted line represents the corresponding reconstructed one. As can be observed, the reconstructed results approach closely the original ones in various cases. To be specific, although there are some slight differences between the compared curves, the main shape tends to be highly similar. Moreover, it should be noted that such results are obtained under the condition of low sampling rate, that is, only 12 bands are used to recover about 200 bands. According to the visual results in Fig. 8, it is indicated that the proposed CACS method can

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effectively recover the spectral curves and, thus, has acceptable performance of spectral fidelity.

For these two images, some experiments are conducted to further demonstrate the effectiveness of the proposed DL and sparse coding operations in improving performance. The detailed experiment configuration is similar to that in Fig. 4. The DCT-CACS and DWT-CACS methods are the variants of the CACS methods, which use the DCT and DWT dictionaries rather than learned dictionary, respectively. In addition, the GSC-CACS method is the CACS method by replacing the general sparse coding method with the proposed sparse coding method. As can be observed in Fig. 9, the proposed DL algorithm helps to significantly improve the reconstruction performance when compared with the DCT-CACS and DWT-CACS methods, in terms of PSNR, SSIM, and SAD. In addition, the proposed modified sparse coding method can also boost the reconstruction performance in some degrees compared with the traditional general ones. This superior performance is in accordance with the results in Fig. 4, further demonstrating the effectiveness of the proposed method in dealing with different HSIs.

V. Conclusion

This article proposes a new CACS method for HSIs. The main contribution lies on incorporating contextual prior into the DL and sparse reconstruction, resulting in a good performance with respect to both the spectral and spatial fidelities. On the one hand, the contextual prior is used for online learning dictionary via jointly representing spectral–spatial correlated pixels within a patch. By this way, the learned dictionary can result in structural and sparser representation of spectral signals, contributing to a better performance of CS reconstruction. On the other hand, the local spatial prior is used in sparse reconstruction by incorporating a weighted smoothing regularization. By solving the optimal problem in the proposed reconstruction model effectively, the homogeneous regions tend to be smooth while the structural details can be preserved well. Moreover, comparison experiments with some state-of-the-art compressive imaging methods can demonstrate the superiority of the proposed methods, in terms of both visual impression and objective metrics.

In this article, we develop an ODL method and a spectral–spatial sparse coding method to improve the reconstruction performance of CS. Some ideas in the proposed method could be further used for classification and anomaly detection with some variants. For instance, the spectral–spatial sparse coding method can produce structural coefficients, which can help to improve the classification accuracy of neighboring pixels. In addition, the model of pursing sparse coefficients during the DL can be used to adaptively identify outliers those are quite different from the majority of neighboring pixels. This model can be extended to solve the anomaly detection problem, considering that searching for some kind of "outliers" is one focused issue in anomaly detection.

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